#HW 1

Graded

Student

Alexis Cortes

Total Points

91 / 100 pts

Question 1

Q1

20 / 20 pts

– 2 pts There is an error in your calculation. Feet should be converted to inches. ft^3 can be converted to inch^3 by multiplying it with 12^3.

✔ **– 0 pts** Click here to replace this description.

– 2 pts Since the Patm is given in lbf so you should get your gauge and system pressure in lbf too. So you should divide rhogh by the gravity.

– 2 pts Absolute or system pressure can be obtained by adding up the gauge pressure with Patm.

– 2 pts The effective height should be 15*sin 40 degrees. You would have got a different answer if you had used 40 degree instead of 30 degree since the question mentioned a 40 degree inclination.

– 5 pts Late

Question 2

Q2

16 / 20 pts

– 0 pts Click here to replace this description.

✔ **– 4 pts** Since, the specific volume has been given, in order to get the density, you need to take the reciprocal of the specific volume.

– 1 pt The answer you got is in Pascal. To convert it to kPa you must divide it by 1000.

– 5 pts Incomplete submission

– 5 pts Late

Question 3

Q3

18 / 20 pts

– 0 pts Click here to replace this description.

✔ **– 2 pts** In the question it has been mentioned that the pressure changes from 1 bar to 200 bar in increments and volume should be calculated next. You had increased volume and then calculated pressure. You are supposed to increase the pressure.

– 2 pts Pressure should be on the y axis and volume should be on the x axis

– 2 pts Please remember to share your data from the next time. The question also mentioned that you should share the screenshot of both the plot and data.

– 2 pts The question said to add 20 increments. So you should have reported 20 data sets.

– 2 pts Please remember to add titles on your axes from the next time.

– 10 pts Incomplete

– 5 pts Late

– 1 pt Add units on axes.

Question 4

Q4

15 / 15 pts

✔ **– 0 pts** Click here to replace this description.

– 2 pts The temperature change is not the same for all. It is the same for C and K. It is the same for F and R. You should clearly state it.

– 2 pts The conversion scale for F to C is wrong. It is (F-32)/1.8 i.e. the entire (F-32) gets multiplied by 5/9 or divided by 9/5.

– 5 pts Incomplete

– 1 pt You should show the conversion of both 70 degree F and 42 degree F. You have not shown conversion for the 70 degree F.

– 5 pts Late

Question 5

Q5

22 / 25 pts

– 0 pts Click here to replace this description.

✔ **– 2 pts** You are supposed to calculate the absolute pressure in 5a. So you add the Patm too for getting that.

✔ **– 1 pt** Pressure inside the submersible is same as the atmospheric pressure.

– 1 pt You should deduct Patm from Pabs to get gauge pressure

– 10 pts Incomplete

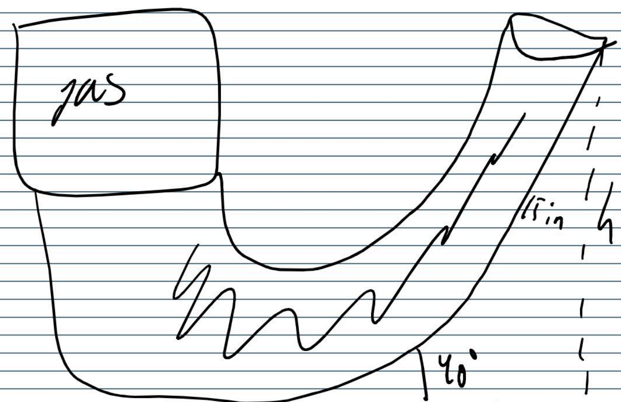
– 10 pts Late

Question assigned to the following page: 1

Q1

Thursday, September 4, 2025 1:18 PM

a.



$P_{atm} = 14.7$ psi

$g = 32.2 \frac{ft}{s^2}$

$\sin 40° = \frac{h}{15 \text{ in}}$

$\rho_{oil} = 54.2 \frac{lb}{ft^3}$

$$P_{gas} = 14.7 \frac{lbf}{in^2} + (\sin 40°)\left(\frac{15}{12} ft\right)\left(54.2 \frac{\cancel{lb}}{ft^3}\right)\left(32.2 \frac{\cancel{ft}}{\cancel{s^2}}\right)\left(\frac{1 \text{ lbf}}{32.2 \frac{\cancel{lb}\,\cancel{ft}}{\cancel{s^2}}}\right)$$

$$= 14.7 \frac{lbf}{in^2} + (\sin(40°))\left(67.75 \frac{lbf}{ft^2}\right)$$

$$= 14.7 \frac{lbf}{in^2} + \left(43.55 \frac{lbf}{ft^2} \cdot \frac{1 \text{ ft}^2}{(12)^2 \text{ in}^2}\right)$$

$$= 14.7 \text{ psi} + 0.3024 \text{ psi} = \boxed{15.0 \text{ psia}}$$

b. $P_g = 15.0 \text{ psia} - 14.7 \text{ psia} = \boxed{0.3024 \text{ psig}}$

c. The inclined apparatus allows for less hydrostatic pressure, making it safer and less prone to accidents. It also makes managing the gas pressure easier, since $P_{hydrostatic}$ is smaller.

Question assigned to the following page: 2

Q2

Friday, September 5, 2025 9:01 AM

$$P_a = 101\ kPa + (0.12 m + h)\left(0.00122 \frac{kg}{m^3}\right)\left(9.81 \frac{m}{s^2}\right)$$

$$P_b = 101\ kPa + (h\ m)\left(0.0122 \frac{kg}{m^3}\right)\left(9.81 \frac{m}{s^2}\right)$$

$$\Delta P \underset{P_a - P_b}{=} (101\ kPa - 101\ kPa) + (0.0014\ Pa + 0.012h\ Pa - 0.012h\ Pa)$$

$$= \boxed{0.0014\ Pa}$$

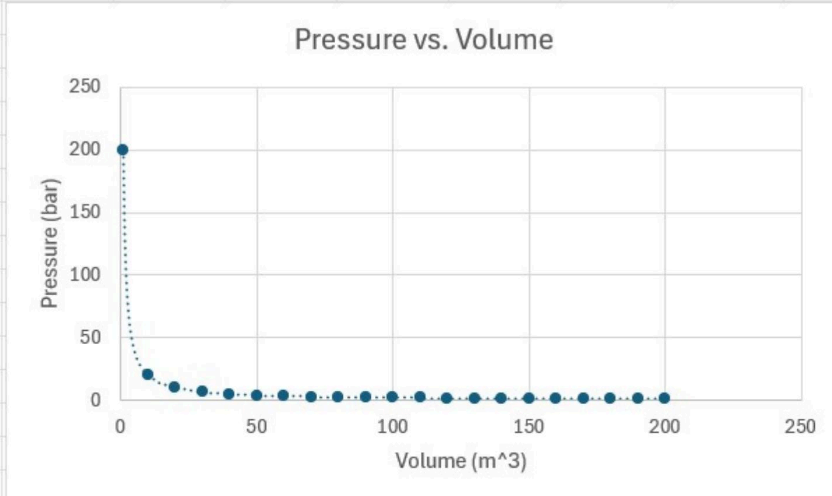
This indicates that the pressure decreases as the pipe's diameter decreases

Question assigned to the following page: 3

Q3

Friday, September 5, 2025 9:01 AM

V (m^3)	P (bar)
1	200
10	20
20	10
30	6.666667
40	5
50	4
60	3.333333
70	2.857143
80	2.5
90	2.222222
100	2
110	1.818182
120	1.666667
130	1.538462
140	1.428571
150	1.333333
160	1.25
170	1.176471
180	1.111111
190	1.052632
200	1



Question assigned to the following page: 4

Q4

Friday, September 5, 2025 9:01 AM

a. $42\,°F + 459.67 = \boxed{501.67\,°R}$

$$\frac{(42\,°F - 32)}{1.8} = \boxed{5.6\,°C}$$

$5.6\,°C + 273.15 = \boxed{278.7\ K}$

$70\,°F + 459.67 = \boxed{529.67\,°R}$

$$\frac{(70\,°F - 32)}{1.8} = \boxed{21\,°C}$$

$21\,°C + 273.15 = \boxed{294\ K}$

Question assigned to the following page: 4

b. $\Delta T_F = 70 - 42 = 28°F$

$\Delta T_C = 21 - 5.6 = 15.4°C$

$\Delta T_R = 529.67 - 501.67 = 28°R$

$\Delta T_K = 294 - 278.7 = 15.3 K$

c. $\Delta T_F = \Delta T_R$

D. $\Delta T_C = \Delta T_K$

$P = h\rho g$

a. $P_{hydrostatic} = (3800\ m)\left(1000\ \frac{kg}{m^3}\right)\left(9.8\ \frac{m}{s^2}\right)\left(\frac{1\ MPa}{1\times10^6\ Pa}\right) = \boxed{37.24\ MPa}$

$= 37.24\ MPa \cdot \frac{1\ atm}{0.101325\ MPa} = \boxed{367.5\ atm}$

$= 37.24\ MPa \cdot \frac{14.50377\ psi}{0.1\ MPa} = \boxed{5401\ psia}$

b. $P_{gauge} = 5401\ psi - 14.7\ psi = \boxed{5386.3\ psig}$ Most loads aren't affected by P_{atm}, so it makes calculations more effective

c. $P_{submersible} = P_{hydro} + P_{atm} = 5401\ psi + 14.7\ psi = \boxed{5415.7\ psi}$

An ideal submersible would balance out the outside pressure by having the exact same pressure inside.

d. An ideal submarine would continuously increase its pressure as it goes deeper, in order to balance out with the external pressure.
It could be done by having it built by a strong material, as well as gas tanks that will increase the pressure inside the submersible.

e. The Titan imploded most likely because it was unable to equilibrate its pressure with the $P_{hydrostatic}$, creating a vacuum into which the $P_{hydrostatic}$ was attracted.

There could have also been failures due to improper structural designs, such as inappropriate building materials or lack of sufficient supports to act against the hydrostatic pressure.